

第二十讲：特征函数

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定义 0.1 若 $X(\omega), Y(\omega)$ 为定义在 Ω 上的实值随机变量, 则称 $Z(\omega) = X(\omega) + iY(\omega)$ 为复随机变量; 称 $\bar{Z} = X(\omega) - iY(\omega)$ 为 $Z(\omega)$ 的复共轭随机变量; 称 $|Z| := \sqrt{X^2 + Y^2}$ 为复随机变量 Z 的模.

定义 0.2 若随机变量 X, Y 的期望 $E(X), E(Y)$ 都存在, 则复随机变量 Z 的数学期望定义为 $E(Z) := E(X) + iE(Y)$.

► $Z_1 = X_1 + iY_1$ 与 $Z_2 = X_2 + iY_2$ 独立当且仅当 (X_1, Y_1) 与 (X_2, Y_2) 独立;



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► $E(e^{iX}) = E(\cos X) + iE(\sin X)$;

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定义 0.3 设 X 是一个随机变量, 称

$$\begin{aligned}\varphi(t) &:= E(e^{itX}) = E[\cos(tX)] + iE[\sin(tX)] \\ &= \int_{-\infty}^{+\infty} \cos(tx) dF(x) + i \int_{-\infty}^{+\infty} \sin(tx) dF(x) \\ &= \int_{-\infty}^{+\infty} e^{itx} dF(x), \quad -\infty < t < +\infty\end{aligned}$$

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若 X 服从正态分布, 即 $X \sim N(\mu, \sigma^2)$, 则 $\varphi(t) = e^{it\mu - \frac{1}{2}\sigma^2 t^2}$.

若 X 服从均匀分布, 即 $X \sim U(a, b)$, 则 $\varphi(t) = \frac{e^{ibt} - e^{ita}}{it(b-a)}$.

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如果 X 为离散型随机变量, 其分布列为 $p_k = P(X=x_k)$, 则

$$\varphi(t) = \sum_{k=-\infty}^{+\infty} e^{itx_k} p_k, \quad -\infty < t < +\infty;$$

如果 X 为连续型随机变量, 其概率密度为 $f(x)$, 则

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2. $\varphi(0) = 1, |\varphi(t)| \leq 1$: $\varphi(0) = E(1) = 1$, 而

$$\begin{aligned} |\varphi(t)| &= |E[\cos(tX)] + iE[\sin(tX)]| = [(E[\cos(tX)])^2 + (E[\sin(tX)])^2]^{1/2} \\ &\leq [E[\cos^2(tX)] + E[\sin^2(tX)]]^{1/2} = 1 \end{aligned}$$

3. $\varphi(-t) = \overline{\varphi(t)}$: $\varphi(-t) = E[\cos(-tX)] + iE[\sin(-tX)] = E[\cos(tX)] - iE[\sin(tX)] = \overline{E[\cos(tX)] + iE[\sin(tX)]} = \overline{\varphi(t)}$. 事实上,

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$$\begin{aligned} \varphi_Y(t) &= E[e^{it(aX+b)}] = E[e^{itax} e^{ibt}] = e^{ibt} E[e^{itax}] \\ &= e^{ibt} E[e^{i(at)x}] = e^{ibt} \varphi_X(at) \end{aligned}$$



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$$\begin{aligned} |\varphi(t)| &= |E[\cos(tX)] + iE[\sin(tX)]| = [(E[\cos(tX)])^2 + (E[\sin(tX)])^2]^{1/2} \\ &\leq [E[\cos^2(tX)] + E[\sin^2(tX)]]^{1/2} = 1; \end{aligned}$$

3. $\varphi(-t) = \overline{\varphi(t)}$: $\varphi(-t) = E[\cos(tX)] - iE[\sin(tX)] = \overline{\varphi(t)}$;
4. 若 $Y = aX + b$, 其中 a, b 为常数, 则 $\varphi_Y(t) = e^{ibt}\varphi_X(at)$, 事实上,

$$\begin{aligned} \varphi_Y(t) &= E[e^{it(aX+b)}] = E[\cos(t(aX+b))] + iE[\sin(t(aX+b))] \\ &= E[\cos(taX)\cos(tb) - \sin(taX)\sin(tb)] + iE[\sin(taX)\cos(tb) + \cos(taX)\sin(tb)] \\ &= \cos(tb)(E[\cos(taX)] + iE[\sin(taX)]) + i\sin(tb)(E[\cos(taX)] + iE[\sin(taX)]) \\ &= e^{itb}Ee^{itaX} = e^{itb}\varphi_X(at) \end{aligned}$$



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5. 若 X, Y 独立, 则 $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$, 事实上,

$$\begin{aligned}\varphi_{X+Y}(t) &= E(e^{it(X+Y)}) = E(e^{itX}e^{itY}) \\&= E[(\cos(tX) + i\sin(tX))(\cos(tY) + i\sin(tY))] \\&= E[(\cos(tX) + i\sin(tX))\cos(tY)] + iE[(\cos(tX) + i\sin(tX))\sin(tY)] \\&= E[\cos(tY)]E[e^{itX}] + iE[\sin(tY)]E[e^{itX}] \\&= E(e^{itX})E(e^{itY})\end{aligned}$$



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6. 更一般的, 若 $X_1, X_2, \dots, X_n$ 相互独立, 则对于 $Y = X_1 + \dots + X_n$ 的特征函数有

$$\varphi_Y(t) = \prod_{k=1}^n \varphi_{X_k}(t)$$



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$$\varphi^{(k)}(t) = i^k E[X^k e^{itX}], \quad \varphi^{(k)}(0) = i^k E[X^k]$$

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8. $\varphi(t)$ 在 $(-\infty, +\infty)$ 上一致连续:

$$\begin{aligned} |\varphi(t+h) - \varphi(t)| &= |E(e^{i(t+h)X}) - E(e^{itX})| = |E[e^{itX}(e^{ihX} - 1)]| \\ &\leq E|e^{itX}(e^{ihX} - 1)| = E|e^{ihX} - 1| = \int_{-\infty}^{+\infty} |e^{ihx} - 1| dF_X(x) \\ &= \int_{|x|>A} |e^{ihx} - 1| dF_X(x) + \int_{-A}^A |e^{ihx} - 1| dF_X(x) \\ &\leq 2 \int_{|x|>A} dF_X(x) + \int_{-A}^A |hx| dF_X(x) \\ &\leq 2P(|X| > A) + hAP(|X| \leq A) \leq 2P(|X| > A) + hA \end{aligned}$$

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9. $\varphi(t)$ 是非负定函数, 即对任意的正整数 n 及 n 个实数 $t_1, \dots, t_n$ 及 n 个复数 $z_1, \dots, z_n$, 均有

$$\sum_{k=1}^n \sum_{j=1}^n \varphi(t_k - t_j) z_k \bar{z}_j \geq 0$$

证明: 由特征函数的定义可得

$$\begin{aligned} \sum_{k=1}^n \sum_{j=1}^n \varphi(t_k - t_j) z_k \bar{z}_j &= \sum_{k=1}^n \sum_{j=1}^n z_k \bar{z}_j E[e^{i(t_k - t_j)X}] = \sum_{k=1}^n \sum_{j=1}^n E[z_k e^{it_k X} \bar{z}_j e^{-it_j X}] \\ &= E\left[\sum_{k=1}^n z_k e^{it_k X} \sum_{j=1}^n \bar{z}_j e^{-it_j X}\right] = E\left[\left|\sum_{k=1}^n z_k e^{it_k X}\right|^2\right] \geq 0 \end{aligned}$$



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常用分布的特征函数 I



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► 指数分布:

$$\varphi(t) = \int_0^{\infty} e^{itx} \lambda e^{-\lambda x} dx = \left(1 - \frac{it}{\lambda}\right)^{-1}$$

- 二项分布: $Y \sim B(n, p)$. 注意到 $Y = X_1 + \cdots + X_n$, 其中 X_i 为独立同分布的随机变量, 且 $X_i \sim B(1, p)$, 故

$$\varphi_Y(t) = \prod_{k=1}^n (pe^{it} + q) = (pe^{it} + q)^n, \quad q = 1 - p$$

- 一般正态分布: $Y \sim N(\mu, \sigma^2)$, 则由 $X := \frac{Y - \mu}{\sigma} \sim N(0, 1)$ 及 $Y = \sigma X + \mu$ 可得

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更进一步, 当 $Y \sim \Gamma(\alpha, \lambda)$, 我们有

$$\varphi_Y(t) = (1 - \frac{it}{\lambda})^{-\alpha}$$

- $\chi^2(n)$ 分布: 因为 $\chi^2(n) = \Gamma(n/2, 1/2)$, 故其特征函数为

$$\varphi(t) = (1 - 2it)^{-n/2}$$



- 伽玛分布: $Y \sim \Gamma(n, \lambda)$. 注意到 $Y = X_1 + \cdots + X_n$, 其中 X_i 独立同指数分布即 $X_i \sim \exp(\lambda)$, 故

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更进一步, 当 $Y \sim \Gamma(\alpha, \lambda)$, 我们有

$$\varphi_Y(t) = (1 - \frac{it}{\lambda})^{-\alpha}$$

- $\chi^2(n)$ 分布: 因为 $\chi^2(n) = \Gamma(n/2, 1/2)$, 故其特征函数为

$$\varphi(t) = (1 - 2it)^{-n/2}$$

例 0.1 利用特征函数的方法求 $\Gamma(\alpha, \lambda)$ 分布的数学期望与方差.

解: Γ 分布的特征函数及其导数分别为

$$\varphi(t) = \left(1 - \frac{it}{\lambda}\right)^{-\alpha}$$

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特征函数与分布函数之间的关系：相互唯一决定



定理 0.1 (逆转公式) 设 $F(x)$ 与 $\varphi(t)$ 分别为随机变量 X 的分布函数和特征函数, 则对 $F(x)$ 的任意两个连续点 $x_1 < x_2$ 有

$$F(x_2) - F(x_1) = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T \frac{e^{-itx_1} - e^{-itx_2}}{it} \varphi(t) dt$$

定理 0.2 (唯一性定理) 随机变量的分布函数由其特征函数唯一决定.

证明: 在逆转公式中令 $x_2 = x, x_1 = y \rightarrow -\infty$ 可得

$$F(x) = \lim_{y \rightarrow -\infty} \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T \frac{e^{-ity} - e^{-itx}}{it} \varphi(t) dt$$

而由分布函数本身是右连续可知, 分布函数任一点的取值可由连续点上的值唯一决定.

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引理 0.1 (狄利克雷积分) 令

$$I(a, T) = \int_0^T \frac{\sin at}{t} dt, \quad \operatorname{sgn}\{a\} := \begin{cases} -1, & a < 0, \\ 0, & a = 0, \\ 1, & a > 0, \end{cases}$$

则

$$\lim_{T \rightarrow +\infty} I(a, T) = \lim_{T \rightarrow +\infty} \int_0^T \frac{\sin at}{t} dt = \frac{\pi}{2} \operatorname{sgn}\{a\}$$

证明: 注意到如果作积分换元 $y = at$ 可得

$$\lim_{T \rightarrow +\infty} I(a, T) = \operatorname{sgn}\{a\} \lim_{T \rightarrow +\infty} I(1, T)$$



只需证明

$$\lim_{T \rightarrow +\infty} I(1, T) = \lim_{T \rightarrow +\infty} \int_0^T \frac{\sin t}{t} dt = \frac{\pi}{2}.$$

注意到 $\frac{1}{t} = \int_0^{+\infty} e^{-ut} du$, 故

$$\begin{aligned} I(1, T) &= \int_0^T \int_0^{+\infty} e^{-ut} \sin t du dt = \int_0^{+\infty} \left[\int_0^T e^{-ut} \sin t dt \right] du \\ &= \int_0^{+\infty} \left[\frac{1}{1+u^2} - \frac{1}{1+u^2} e^{-uT} (u \sin T + \cos T) \right] du \\ &= \frac{\pi}{2} - \int_0^{+\infty} \frac{u \sin T + \cos T}{1+u^2} e^{-uT} du \rightarrow \frac{\pi}{2} \end{aligned}$$



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逆转公式的证明



证明: 考虑积分

$$\begin{aligned} J(T) &= \frac{1}{2\pi} \int_{-T}^T \frac{e^{-itx_1} - e^{-itx_2}}{it} \varphi(t) dt \\ &= \frac{1}{2\pi} \int_{-T}^T \left[\int_{-\infty}^{+\infty} \frac{e^{-itx_1} - e^{-itx_2}}{it} e^{itx} dF(x) \right] dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\int_{-T}^T \frac{e^{it(x-x_1)} - e^{it(x-x_2)}}{it} dt \right] dF(x) \\ &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \left[\int_0^T \frac{\sin t(x-x_1)}{t} dt - \int_0^T \frac{\sin t(x-x_2)}{t} dt \right] dF(x) \\ &= \frac{1}{\pi} \int_{-\infty}^{+\infty} [I(x-x_1, T) - I(x-x_2, T)] dF(x) \\ &\rightarrow \frac{1}{\pi} \int_{-\infty}^{+\infty} [\operatorname{sgn}\{x-x_1\} - \operatorname{sgn}\{x-x_2\}] dF(x) \quad T \rightarrow +\infty \\ &= \int_{x_1}^{x_2} dF(x) = F(x_2) - F(x_1) \end{aligned}$$

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J(T) &= \frac{1}{2\pi} \int_{-T}^T \frac{e^{-itx_1} - e^{-itx_2}}{it} \varphi(t) dt \\
&= \frac{1}{2\pi} \int_{-T}^T \left[\int_{-\infty}^{+\infty} \frac{e^{-itx_1} - e^{-itx_2}}{it} e^{itx} dF(x) \right] dt \\
&= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\int_{-T}^T \frac{e^{it(x-x_1)} - e^{it(x-x_2)}}{it} dt \right] dF(x) \\
&= \frac{1}{\pi} \int_{-\infty}^{+\infty} \left[\int_0^T \frac{\sin t(x-x_1)}{t} dt - \int_0^T \frac{\sin t(x-x_2)}{t} dt \right] dF(x) \\
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定理 0.3 (逆转公式一般情形) 设 $F(x)$ 与 $\varphi(t)$ 分别为随机变量 X 的分布函数和特征函数, 则对任意的 $x_1, x_2 \in R$ 有

$$\frac{F(x_2 - 0) + F(x_2)}{2} - \frac{F(x_1 - 0) + F(x_1)}{2} = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T \frac{e^{-itx_1} - e^{-itx_2}}{it} \varphi(t) dt$$

连续型分布情形下的逆转公式



定理 0.4 若 X 为连续型随机变量, 其密度函数为 $p(x)$, 其特征函数为

$$\varphi(t) = \int_{-\infty}^{+\infty} e^{itx} p(x) dx,$$

如果 $\int_{-\infty}^{+\infty} |\varphi(t)| dt < +\infty$, 则

$$p(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-itx} \varphi(t) dt$$

证明: 由分布密度的定义可知

$$\begin{aligned} p(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{e^{-itx} - e^{-it(x+h)}}{it \cdot h} \varphi(t) dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \lim_{h \rightarrow 0} \frac{e^{-itx} - 1}{-it \cdot h} e^{-itx} \varphi(t) dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-itx} \varphi(t) dt \end{aligned}$$

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